Fear is contagious too: behavioural feedback in epidemic spreading on networks

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Abstract

We couple a network SIR epidemic with a second contagion, risk perception, and compare three behavioural-feedback rules head to head: local observation of sick contacts, social imitation of cautious neighbours, and global broadcast nudges (uniform or hub-targeted). Across 13,290 simulations on Watts–Strogatz, Barabasi–Albert and degree-matched Erdos–Renyi networks ($N = 1000$), local observation averts the most infections per unit of social cost when averaged over the tested grid on each topology. Running to extinction removes broadcast's apparent final-size advantage on Watts–Strogatz networks. Degree-aware contact costs remove most of hub targeting's efficiency advantage, while intermediate fear decay produces repeated prevalence peaks under broadcast. These are comparative findings within synthetic networks, not calibrated intervention recommendations.

1 Introduction

Epidemics spread over contact networks, but so does fear of infection, and fear changes contacts [4]. Classical compartmental models [1] and their network extensions [5] usually treat behaviour as fixed, even though real responses (avoiding contacts, masking, staying home) feed back on transmission. Our research question is: which behavioural-feedback rule minimises epidemic final size *per unit of social cost*, measured as contacts forgone? We test three candidate theories in one framework: (i) imitating neighbours' precautions, (ii) local observation of sick contacts, and (iii) broadcast nudges triggered at a prevalence threshold.

Our proposal stated three hypotheses:

- **H1:** intermediate fear-memory decay produces multi-wave dynamics.
- **H2:** on scale-free networks hub-targeted nudges dominate, while on homogeneous networks they do not.
- **H3:** imitation-based behaviour amplifies and overshoots, producing oscillations.

Coupled awareness and disease models already show that local protective responses can reduce epidemic spread [6]. Our contribution is a controlled comparison of three response rules on the same epidemic engine and three topologies, with explicit contact-cost accounting. We investigate how memory and targeting affect this comparison; the hub-pool ceiling is an imposed policy constraint, not a new network law.

2 Model

Each node carries a disease state S, I or R and a fear level $f \in [0, 1]$. A node is *cautious* when $f > \theta$ with $\theta = 0.5$. Updates are synchronous with fixed parameters $N = 1000$, per-edge transmission probability $\beta = 0.12$, recovery probability $\gamma = 0.05$, caution effectiveness $q = 0.8$, broadcast trigger prevalence 0.02, hub pool 10% by degree, and 10 initially infected nodes. Topologies are Watts–Strogatz (WS, $k = 6$, rewiring 0.1), Barabasi–Albert (BA, $m = 3$) [2, 3], and Erdos–Renyi (ER) with $p = 6/(n - 1)$ so the expected mean degree matches at 6.

Fear updates. Let I_j indicate infected neighbours of i, C_i the cautious-neighbour fraction and $\hat{I}_i$ the infected-neighbour fraction. Observation adds fear from sick contacts and forgets geometrically: $f_i \leftarrow f_i + \alpha\hat{I}_i - \delta f_i$. Imitation copies caution while retaining a local risk cue, with 2% innovators seeded cautious so copying has something to amplify: $f_i \leftarrow f_i + \alpha(C_i - f_i) + \alpha\hat{I}_i$, then multiplied by $(1 - \delta/2)$. Thus imitation has slower effective forgetting at the same nominal δ; this is a modelling assumption, not a matched-memory comparison. Cautious flags are initialised from the seeded fear levels, and C_i uses the previous step's flags. Broadcast decays fear by $(1 - \delta)$ and, when prevalence reaches 0.02, sets fear to 1 for compliers, each included with probability p_{comply} (uniform) or, for hub targeting, restricted to the top degree decile with probability $\min(1, \text{budget}/0.1)$ so budgets up to 10% reach equal expected node counts per triggered update under both policies. Uniform sampling can reach new nodes on successive updates, whereas the hub policy repeatedly samples its fixed pool. Here α is coupling strength for local rules and the compliance budget for broadcast, and δ is fear-memory decay. All fears are clipped to $[0, 1]$.

Disease updates. A susceptible node with c_1 non-cautious and c_2 cautious infected neighbours escapes with probability $e_1^{c_1} e_2^{c_2}$, where $e_1 = 1 - \beta(1 - qc_i)$ and $e_2 = 1 - \beta(1 - q)(1 - qc_i)$ with $c_i \in \{0, 1\}$ its own caution state, so caution protects both ways. Recovery and infection both use the disease state at the start of a step, so an infector can transmit and recover during the same update. Infected nodes recover with probability γ. Outcomes are final size (fraction ever infected), peak prevalence, duration, detected wave count (peak detection with prominence $\max(0.01, 0.2 \times \text{peak})$ and at least 10 steps separation), oscillation index, and social cost. Our primary degree-aware cost is $K_D = (q/2) \sum_t \sum_i c_i(t) k_i$; the original mean-degree proxy $K_M = (q\bar{k}/2) \sum_t \sum_i c_i(t)$ is retained for sensitivity analysis. Here $c_i(t)$ is the caution applied during that step's transmission. These are protective-effort proxies in edge-steps: adjacency is fixed, edges are not deleted, and steps are not calibrated to days. The 1/2 factor is an endpoint-accounting convention, not an exact count of removed edges. Costs end at disease extinction and exclude any later residual caution. Efficiency is $N(F_0 - F)/K$, infections averted per edge-step, with a negligible denominator offset of one edge-step to handle zero cost. Runs stop at extinction, with a 5,000-step safety limit; unfinished runs are flagged and cannot supply a final size to the experiment pipeline. The homogeneous ODE is a reference approximation, not an exact validation target for a clustered, discrete-time network.

3 Experimental design

Three regenerated grids total 13,290 configurations: a full topology×rule×α×δ pilot (9,090 runs, 30 seeds), an equal-budget hub-versus-uniform zoom (3,600 runs, 30 seeds), and a transmission sweep over $\beta \in \{0.03, 0.06, 0.09, 0.12, 0.2\}$ (600 runs, 20 seeds), plus 80 WS runs against a homogeneous SIR ordinary differential equation. Measures, seeds, and the no-behaviour baseline are fixed across comparisons. The pilot uses $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ and $\delta \in \{0, 0.05, 0.15, 0.3, 0.6\}$;

the zoom uses budgets $\{0.02, 0.05, 0.08, 0.10, 0.15\}$ and $\delta \in \{0, 0.05, 0.15, 0.3\}$. Seeds are 0–29, or 0–19 for transmission and ODE comparisons. The transmission comparison fixes observation at $\alpha = 0.75$, $\delta = 0.10$. Means aggregate 600 runs per positive-coupling tradeoff point and 120 per hub-budget point. Efficiency intervals use 2,000 paired bootstrap resamples of seed clusters, preserving each seed's baseline and parameter cells; transmission and ODE error bars show sample SD across 20 seeds. Boundary checks cover zero transmission, immediate recovery when $\gamma = 1$ and $\beta = 0$, zero initial infections, determinism, and the no-behaviour limit. Regression tests cover termination flags, cost timing, innovator initialisation, degree accounting, sparse/dense trajectory equivalence, and full-configuration caching. Cache identities include every simulation parameter and a fingerprint of the model source. `python -m src.run_sweeps --workers 4` regenerates all grids; `notebooks/04_analysis.ipynb` regenerates the figures and summaries. `data/` holds the corrected CSVs and provenance; earlier outputs are retained as legacy data and are not used in the analysis: `https://github.com/idawang142/4403PROJECT`.

4 Results

Observation has the highest grid-average efficiency. Under degree-aware accounting, observation averts 0.00498 infections per edge-step on WS (95% bootstrap interval 0.00478–0.00519), 0.00236 on BA (0.00229–0.00243), and 0.00209 on ER (0.00200–0.00218; Fig. 1, Table 1). It also leads under the mean-degree proxy. This is a ratio of grid-average benefit to cost, not a claim that observation wins every parameter cell. On WS, observation's mean final size is 0.603 versus 0.642 for uniform broadcast, which costs 848,000 versus 80,000 edge-steps, about 10.6 times as much. Uniform broadcast has smaller grid-average epidemics on BA and ER, but higher costs. All runs reach extinction; the longest lasts 1,197 steps. The earlier 500-step cutoff had made broadcast look more effective by excluding later infections and protective effort.

Table 1: Grid-average outcomes at extinction (600 runs per row). Cost is the degree-aware proxy in edge-steps; efficiency is infections averted per edge-step, shown in units of 10^{-3}.

Topology	Rule	Final size	Degree-aware cost	Efficiency ($\times 10^{-3}$)
WS	observation	0.603	79,696	4.98
BA	observation	0.660	140,227	2.36
ER	observation	0.695	140,046	2.09
WS	imitation	0.657	251,777	1.36
BA	imitation	0.651	308,351	1.10
ER	imitation	0.687	331,507	0.91
BA	broadcast	0.586	635,082	0.64
BA	broadcast_hub	0.944	93,962	0.50
WS	broadcast	0.642	847,806	0.42
WS	broadcast_hub	0.985	34,700	0.42
ER	broadcast	0.666	828,287	0.39
ER	broadcast_hub	0.977	45,064	0.24

Memory governs regimes; broadcast chatters (H1 reframed). Final size under full observation coupling rises from 0.073 at $\delta = 0$ to 0.961 at $\delta = 0.6$ on WS (Fig. 2): memory, not just responsiveness, protects. Broadcast shows the sharpest regime change: with perfect memory ($\delta = 0$) prevalence is held down in about one wave, while small positive decay ($\delta = 0.05$) lets protection lapse and reignite, averaging 19.10 detected peaks with a maximum of 35 over the positive-compliance grid, compared with 1.04 at zero decay. The largest mean peak count occurs at $\delta = 0.05$, supporting H1 for this broadcast grid. These algorithmic peaks need not be independent outbreaks, and this finite sweep does not establish a phase transition.

Hub efficiency depends on contact accounting (H2 not general). The mean-degree proxy gives hub targeting higher BA efficiency at every tested budget. Degree weighting removes most of this advantage: the hub point estimate exceeds uniform only at budget 0.02 (0.000330 versus 0.000289); uniform is more efficient at 0.05–0.15. At 0.15, the corresponding efficiencies are 0.000480 for hubs and 0.001163 for uniform (Fig. 3). Thus H2 is not generally supported by the primary cost measure. Hub final size plateaus near 0.944 from budget 0.10, while uniform improves from 0.809 to 0.723. The plateau follows the imposed pool cap; above 10%, per-trigger coverage is no longer equal. Targeting has smaller raw effects on WS and ER under this policy, rather than being intrinsically ineffective on those networks.

The variability index cannot establish overshoot (H3 unresolved). Imitation has the highest oscillation index among the rules (1.367), but the no-behaviour baseline scores higher still (1.456) because the index rewards a single tall peak as well as genuine fluctuation, so H3 remains unresolved by this metric. A fluctuation-specific measure or trajectory analysis is needed before claiming overshoot.

Observation remains effective against its baseline. Without behaviour all topologies saturate near complete infection by $\beta = 0.09$; observation at one fixed untuned setting ($\alpha = 0.75$, $\delta = 0.10$) holds WS below 0.6 even at $\beta = 0.20$, and WS is the most suppressible topology at every tested β (Fig. 4). The comparison includes no other behavioural rules, so it does not establish their ordering across transmission rates. Against the homogeneous reference, at $\beta = 0.03$ WS infects 75.1% (SD 7.5pp) against 97.0% for the ODE (Fig. 6). Clustering and discrete-time effects are plausible contributors to this gap, but their contributions have not been isolated; this is a reference comparison rather than proof of engine validity.

5 Discussion and conclusions

Local observation has the highest grid-average efficiency under both cost proxies. A plausible explanation is that local risk links protection to nearby infection; this mechanism has not been isolated by an ablation. Broadcast generates repeated threshold responses and persistent infections, making extinction-complete evaluation essential. Hub targeting illustrates how a seemingly favourable efficiency ranking can depend on underpricing high-degree nodes. The evidence supports a model comparison, not advice for real populations.

All networks are synthetic, costs are endpoint-effort proxies, and residual caution after extinction is excluded. Rankings depend on grid weights and the cost convention. The fixed trigger and pool cap constrain interventions; nominal decay does not match effective memory across rules. Wave counts depend on the detector, while the variability index leaves H3 unresolved. Bootstrap intervals capture seed variability within the fixed design, not uncertainty in these assumptions. Useful extensions are empirical contact networks, matched effective memory, adaptive triggers, wave-detector sensitivity, and cost-matched policy frontiers.

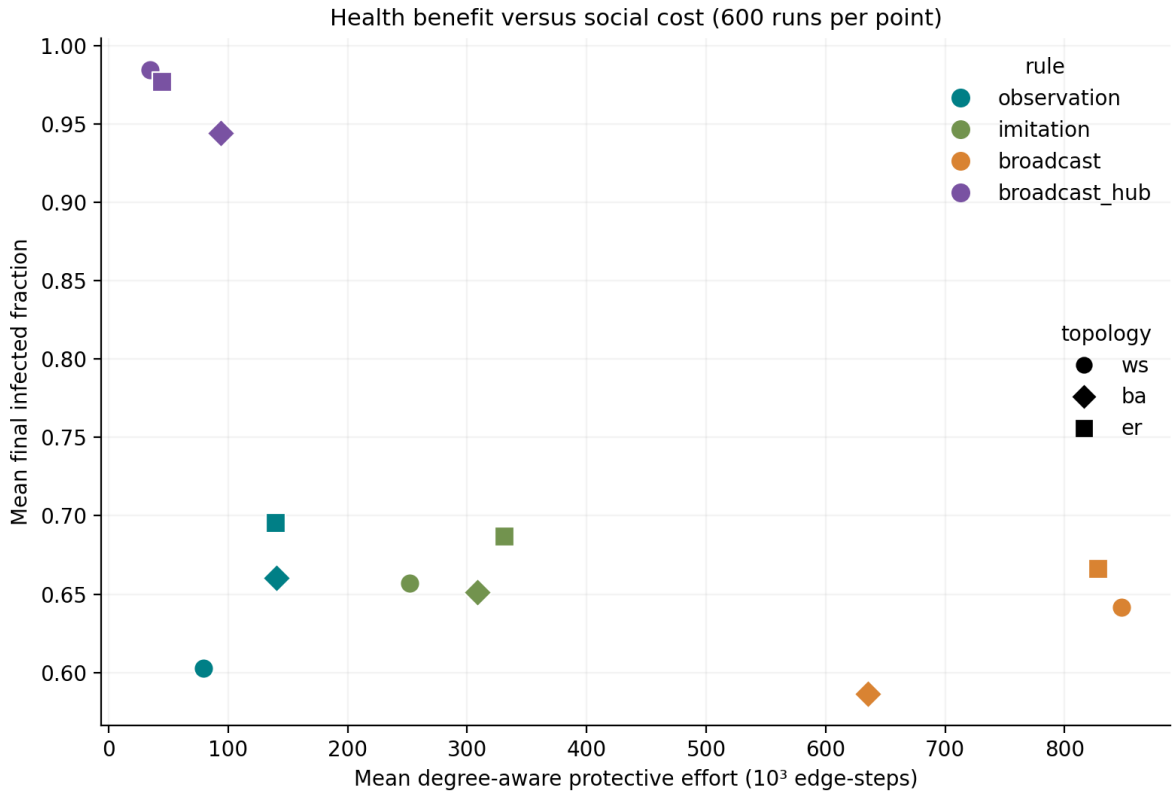


Figure 1: Health benefit versus social cost selects observation. Mean final infected fraction against mean degree-aware protective effort in thousands of edge-steps; each point averages 600 runs. Colour marks the rule, symbol the topology. Lower and further left is better on both axes.

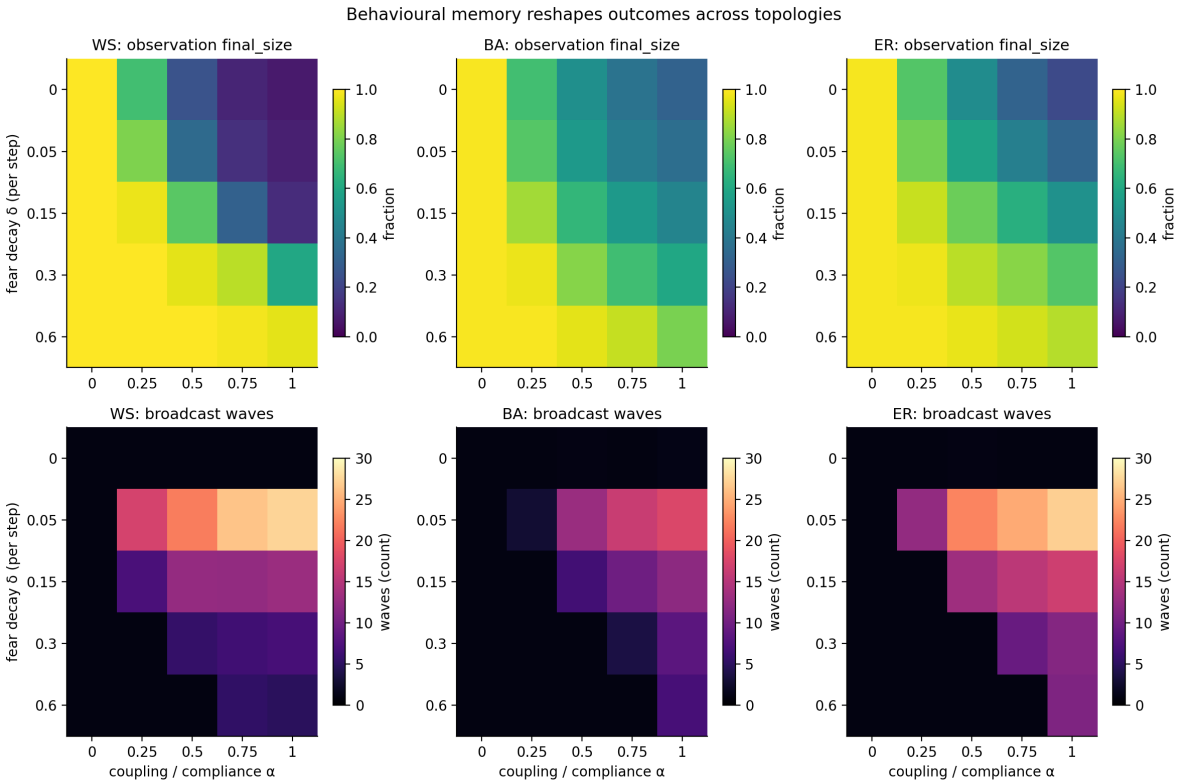


Figure 2: Behavioural memory reshapes outcomes. Top: mean final infected fraction under local observation. Bottom: mean detected wave count under global broadcast. Each cell averages 30 seeds; colour scales are fixed across topologies within a row (fraction 0 to 1; detected peaks 0 to 30).

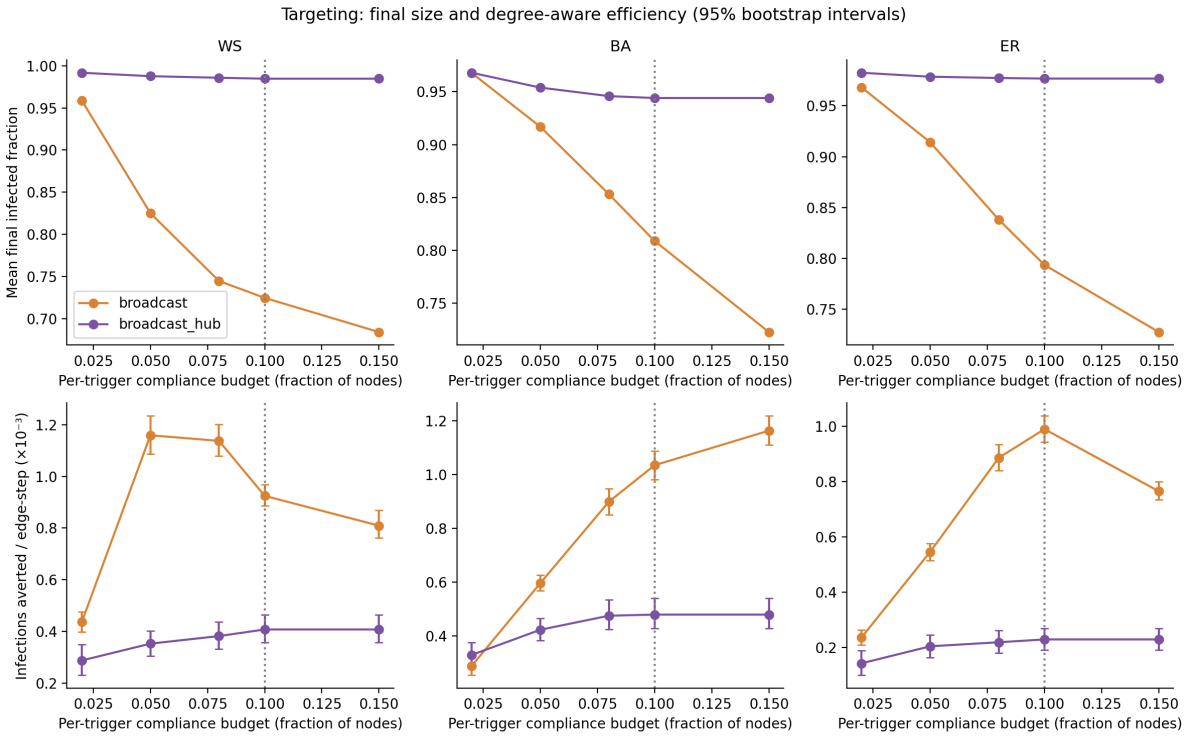


Figure 3: Targeting depends on the cost measure. Top: mean final infected fraction. Bottom: degree-aware efficiency with 95% paired seed-bootstrap intervals. Each point pools 120 runs; budget is the expected fraction of recipients per triggered update. The dotted line marks the 10% pool limit.

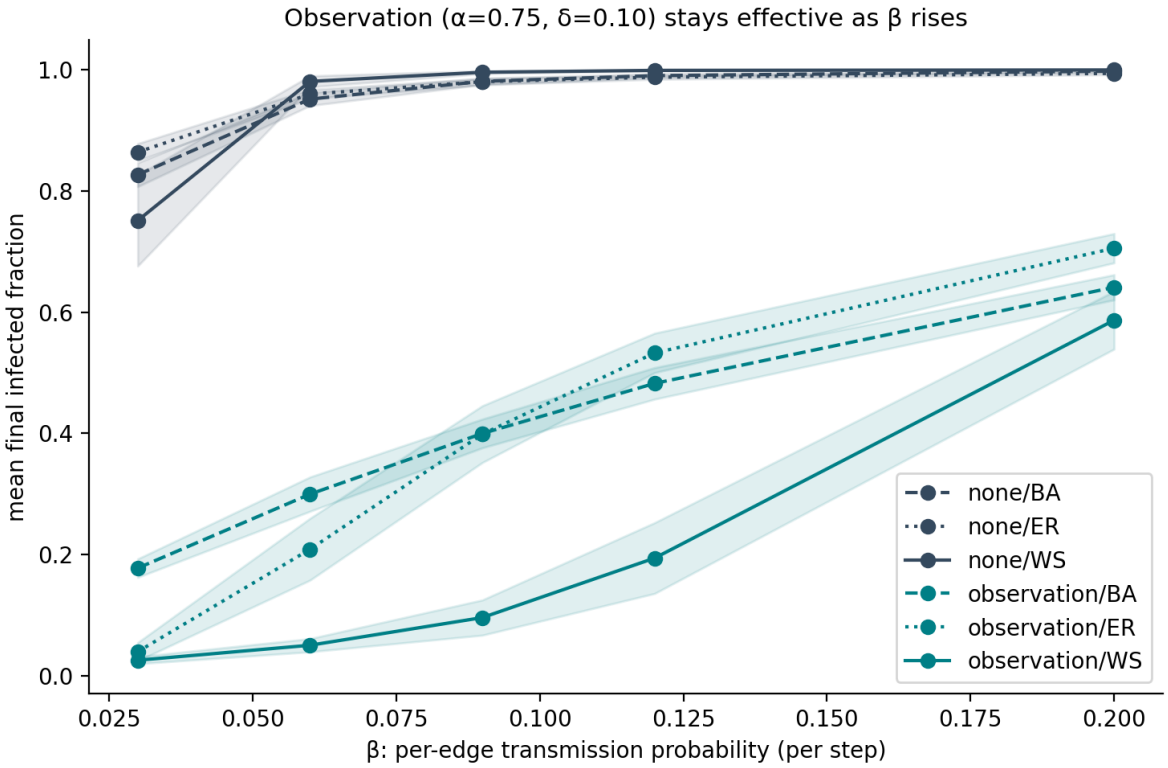


Figure 4: Observation stays effective as transmission rises. Mean final infected fraction against per-edge transmission probability per step for no behaviour and observation ($\alpha = 0.75$, $\delta = 0.10$); bands show ± 1 sample SD across 20 seeds. Line style marks topology, colour marks rule.

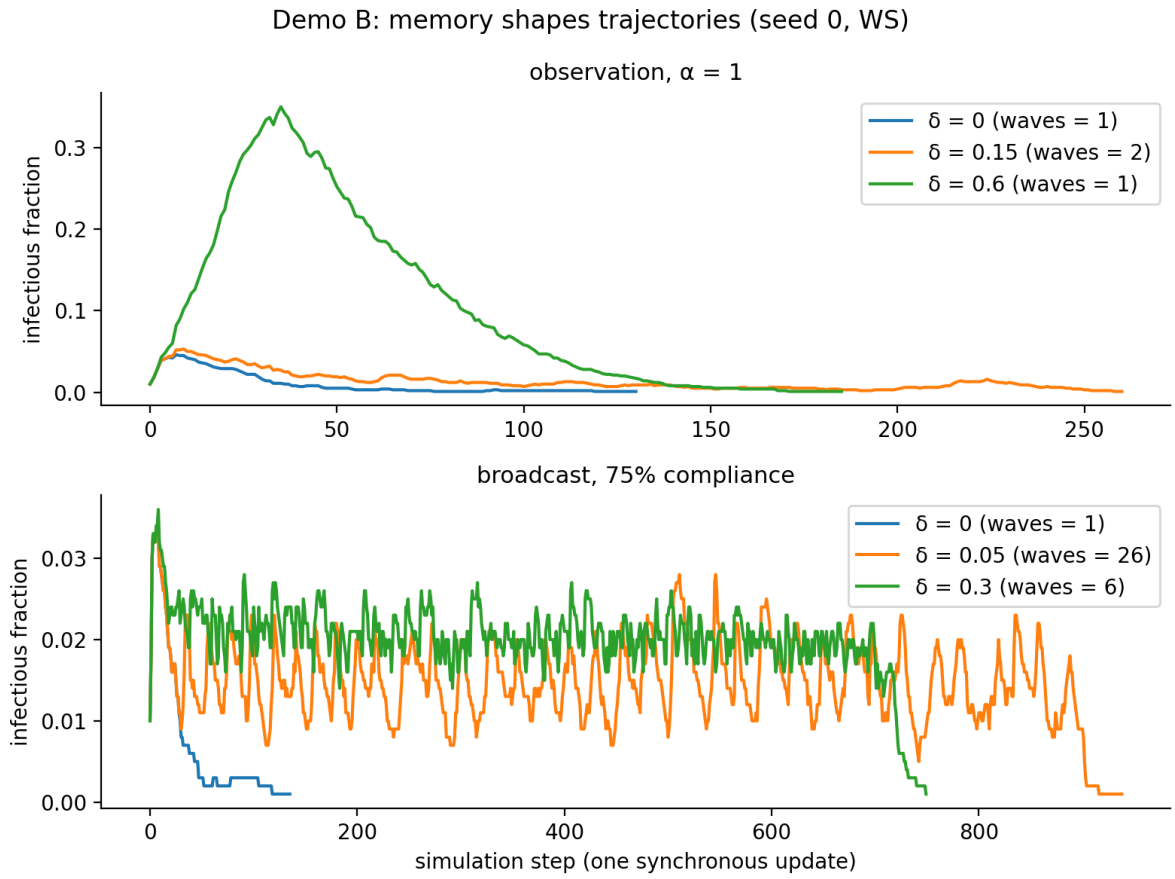


Figure 5: Memory shapes trajectories. Infectious fraction against simulation step for single WS runs (seed 0): local observation at full coupling (top) and 75%-compliance broadcast with trigger 0.02 (bottom) across fear decay rates. Every illustrated run ends at extinction.

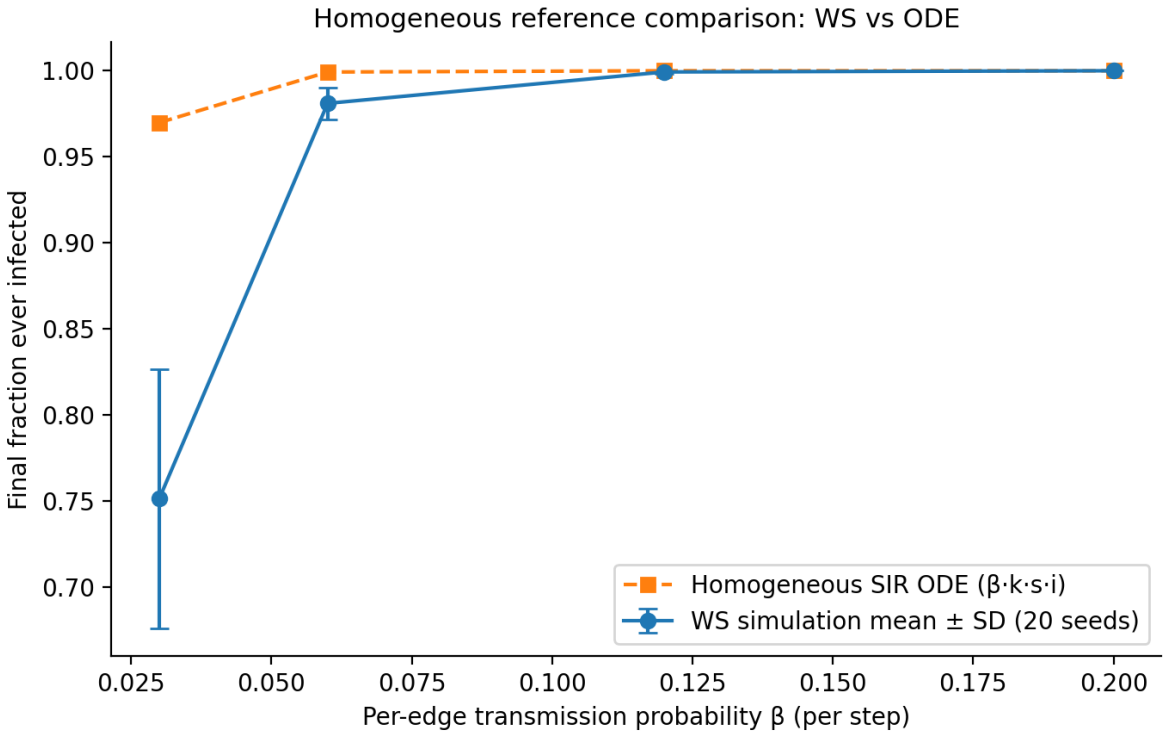


Figure 6: Homogeneous reference comparison. WS mean final size with sample SD (20 seeds) against the SIR ordinary differential equation with incidence βksi ($k = 6$) at $\gamma = 0.05$. The gap at low β is reported without attributing it to a single mechanism.

References

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